

GANGHUA WANG

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Address: 492 Ford Hall, 224 Church Street SE
Minneapolis, MN 55455

EDUCATION

University of Minnesota, Twin Cities

Ph.D. in Statistics

Aug. 2019 - present

Minneapolis, MN

Advised by [Prof. Jie Ding](#) and co-advised by [Prof. Yuhong Yang](#)

Peking University

B.S. in Statistics, Minor in Economics

Sept. 2015 - July 2019

Beijing, China

EXPERIENCE

Cisco Research Fellow: Privacy-preserving machine learning

Jan. 2021 - present

Supervised by Prof. Jie Ding and Prof. Yuhong Yang, University of Minnesota

Collaborated with the Cisco research team

Minneapolis, MN

We formalized the definition of model privacy and proposed model defense mechanisms that provide rigorous guarantees.

Research assistant: Deep learning theory

Sept. 2021 - Present

Supervised by Prof. Jie Ding, University of Minnesota

Minneapolis, MN

Research assistant: Data privacy

June 2020 - Aug. 2020

Supervised by Prof. Jie Ding, University of Minnesota

Minneapolis, MN

We proposed a novel framework to protect data privacy for each individual in the data collection procedure, which is different from differential privacy.

Project: Travelers Modeling Problem

Nov. 2019 - Dec. 2019

Peace of Mind Insurance Company

Minneapolis, MN

A competition to predict future policies based on the historical insurance data, held in [Kaggle](#). Collaborated with other graduate students, we ranked first in the public leaderboard, and second in the private leaderboard.

Summer internship: Component analysis in brain image

July. 2018 - Sept. 2018

Supervised by Prof. Hongyu Zhao, Yale University

New Haven, CT

We proposed to extract functional brain neural network components using the Latent Dirichlet Allocation method, which facilitates the understanding of the intrinsic structure of the human brain.

Research assistant: Community detection with co-variates

Dec. 2017 - June 2018

Supervised by Prof. Jinzhu Jia, Peking University

Beijing, China

We adopted a maximum likelihood estimation-based method to combine the information from both co-variates and the network to detect communities under the stochastic block model.

PUBLICATIONS

Journal

[†] indicates corresponding author(s)

- [1] **Ganghua Wang**, Jie Ding, and Yuhong Yang[†]. “Regression with Set-Valued Categorical Predictors”. In: *Statistica Sinica, to appear* (2021).

Submitted

- [2] Gen Li, **Ganghua Wang**, Yuantao Gu, and Jie Ding[†]. “Provable Identifiability of ReLU Neural Networks via LASSO Regularization”. In: *arXiv preprint* (2021).
[3] **Ganghua Wang** and Jie Ding[†]. “Subset Privacy: Draw from an Obfuscated Urn”. In: *arXiv preprint* (2021).

HONORS AND AWARDS

Awarded by University of Minnesota, Twin Cities

Summer Research Fellowship 2020

School of Statistics First Year Scholarship 2019

Awarded by Peking University

The Academic Excellence Scholarship (3 times) 2016 - 2018

Fang Zheng Scholarship 2017

Wu Si Scholarship 2016, 2018

Freshman Scholarship 2015

TEACHING

Teaching Assistant, University of Minnesota, Twin Cities

STAT 4102 Theory of Statistics II Fall 2020

STAT 3021 Introduction to Probability and Statistics Spring 2020

STAT 3011 Introduction to Statistical Analysis Fall 2019

SERVICES

Reviewer of

The International Conference on Acoustics, Speech, & Signal Processing (ICASSP) 2022

TECHNICAL STRENGTHS

Computer Languages Python, MATLAB, R, SQL